

Assessment 4	Mobile application development
Due date	See Submission Link for details
Weighting	50%
Word count/length	2000 words equivalent (+/-10%; The reference list is not included in the word count but in-text citations are)
SILOs	- Analyse the principles underlying mobile applications and mobile computing by comparing them to applications on different platforms (SILO 1). - Investigate mobile user interface abstractions and mobile usability issues, design and program mobile applications for mobile devices on at least one platform (SILO 2). - Design and construct applications based on the notion of context-aware computing, including using sensors and location services (SILO 3). - Utilize and implement software development methodologies to build and enhance modern mobile applications in teams (SILO 4).
Graduate capabilities	- Communication - Inquiry and Analysis - Personal and Professional - Discipline Knowledge and Skills
Generative AI use	The applicable AI use category is: Full AI The AI Assessment Scale resource provides guidance on the appropriate use of GenAI tools in assessments and explains what is expected in each category.
Feedback	Written feedback will be available within three weeks of the due date.
Short extension	Eligible: This assessment is eligible for a short extension. To apply, visit Assessment 4: 3-day short extension and add yourself to the extension group.

Purpose

In this assessment, you will design, build, test, and present a complete mobile application. Development will be carried out across three Agile-style sprints, demonstrating progressive feature delivery, collaboration, and professional development practices.

Task details

Working in teams (teams of two/pairs), you will **implement a complete mobile application** over three Agile-style sprints. The task includes software development, testing and deployment, documentation, and a pitch-style presentation of the final product

Instructions

Both students need to submit. In case of unequal contribution by team members markers will examine contributions.

You must complete the project in at least **three development sprints**:

- Sprint 1: Weeks 3–5
- Sprint 2: Weeks 6–8
- Sprint 3: Weeks 9–11

Each sprint must demonstrate:

- Clearly defined sprint goals and user stories (documented in **Azure DevOps** or a similar tool)
 - Appropriate task decomposition and allocation
 - Incremental feature delivery
 - Regular and meaningful **GitHub commits**, linked to sprint tasks or user stories
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Implementation requirements

Your completed application must include:

- Functional screens with clear navigation
 - Integration of **Firebase** (Firestore, Authentication, Test Lab)
 - Use of **device capabilities**, such as:
 - Sensors (e.g. accelerometer, gyroscope, torch), and/or
 - Maps and GPS
 - Screens (all) + data between screens
 - Battery
 - Parallel Programming
 - Work Manager/Task Manager
 - Notification
 - AdMods
 - Testing (Test Lab + Jest)
 - APK and Builds
 - Reliable data storage and retrieval (SQLite or Relational Database)
 - A clear explanation of:
 - The three Firebase technologies used
 - Why each technology was selected
 - Reflection on limitations and future improvements
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Software implementation expectations

Your codebase should demonstrate:

- A clear and structured application architecture
 - Logical separation of components, services, and screens
 - Clean, readable, and well-organised code
 - No hard-coded credentials or secrets (e.g. API keys in source files)
 - Appropriate use of configuration and environment variables
 - Evidence of collaboration through GitHub commit history from all team members
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User manual (~3 pages per person)

Each student must contribute to a user manual that includes:

- Getting started instructions
- How to use the application

- Common issues and troubleshooting tips

Testing and deployment report (~3 pages per person)

You must provide evidence of **comprehensive testing**, including:

- Unit and integration and End-to-End testing (e.g. **Jest** – at least one unit, end to end and integration test per person)
 - Test descriptions and results
- Use of **Firestore Test Lab (one different per person)**
 - Describe Devices used + Test
 - Discussion of limitations and implications of automated testing
- Reflection on:
 - What worked well
 - Issues discovered
 - How testing informed improvements

Screenshots, logs, and/or short videos should be used as supporting evidence.

Pitch presentation

Each student must submit a **5-minute pitch presentation** introducing the application as a potential product.

The presentation must include:

- The problem the app addresses
- Demonstration of key features
- Overview of testing methods and results
- Lessons learned during development
- Reflection on future enhancements
- Brief Q&A-style reflection (anticipated questions and responses)

Submission requirements

You must submit:

- Zipped source code (excluding credentials and secrets)
- GitHub repository link
 - Screenshots showing individual contributions (minimum one per branch or contributor)
- Azure DevOps (or similar) screenshots
 - Sprint boards
 - User stories
 - Testing tasks
- MS Teams collaboration evidence (where applicable)
- User manual (~3 pages per person)
- Testing and deployment report (~3 pages per person)
- Screenshots or videos showing:
 - Application functionality
 - Testing results (including Firestore Test Lab)
- Presentation Slides (if using them – avoid using them – show the app instead)

You should expect to spend approximately **50–60 hours** completing this task.

Suggested Schedule:

Module	Assessment Implementation
Introduction to Mobile Applications	Understand the User Specification

User Interface and useability Mobile Operating Systems	Plan out the Development
Planning, Navigation and wireframes Mobile Application Framework	Create the screens and navigation
Intent and Broadcast Messages Threads, Handlers, Background (parallel programming)	Parallel Programming (Background tasks)
Notification Work Managers Fragment and Passing Variables between screens	Notification Work Managers Data passing between screens
Content Providers: SQLiteDB, Firebase, Location	SQLite Firebase: Authentication + Firestore
Location Services and Battery	GPS + Maps Reviews Battery
Gesture, Sensors, Camera, Connectivity	Camera Accelerometer Gyroscope
Biometrics, APIs, AdMob, Theme, Updates	AdMob Themes
Testing	Testing (Jest x3 each Unit & Integration + E2E + Firebase Test Lab)
<i>Mobile Security and Builds</i>	Build APK + Data Encryption

Resources and readings relevant to the assessment

N/A

Assessment criteria

[Submission Link with marking criteria](#)

Submission details

[Submission Link](#)

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[← Joining/Forming Teams \(Bundoora\)](#)

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